

1. * We need a no. that keeps changing.

→ start with ~~initializing~~ initializing, $i = 0$.

→ Then repeatedly; check if i is ≤ 10

↓
print it

↓
increase it by 1.

so,

0 → print → 1

1 → print → 2

2 → print → 3

$i = 0$

while [$i \leq 10$]

do

echo i

$i = i + 1$

2. → Start with $i = 0$

→ Check if the number can be divided by 2 without a remainder.

→ If yes, then write down the number.

→ If no, do not write the number.

→ add 1 to the no.

→ Repeat the same process till you reach 50.

~~Start~~ { Set $i = 0$

while [$i \leq 50$]

{ if [$i \% 2 == 0$]

print i

}

set $i = i + 1$

3.) We'll take user input
take a number $\downarrow$ and call it n
start another variable $i = 1$.

$\downarrow$
Multiply $n \times i$.

$\downarrow$
Print the answer

$\downarrow$
Repeat until $i = 15$.

```
echo "Enter a number "
```

```
read n
```

```
i = 1
```

```
until i > 15
```

```
do
```

```
{ echo "$n x $i = $(($n*$i)) "
```

```
}
```

```
done.
```

4.) read -ra numbers < num.txt

~~read -ra numbers~~

file where
variables are
stored

File which
has the number.

file where we
have to put
numbers into
array

→ Read the numbers
→ store the in array
→ Pick the first number
→ multiply it by 2.
~~then~~
print $2 * x$

→ Repeat until there are no
no. left.

5.) take user ~~output~~ input

↓
call them a & b
↓

First check whether $b \neq 0$

↓
If $b = 0$, say Division is not possible.

↓
else, calculate:

- $a/b \rightarrow$ quotient
- $a \% b \rightarrow$ remainder

↓
print both

~~6.1.1~~ 6.) Take two numbers

↓
compare them

↓
If first $>$ second $\rightarrow$ first is maximum

↓
else, $\rightarrow$ second is max.

↓
print max

7. take user input (folder name)

↓
check whether folder exists

↓
If it exists:

- create the folder

else; tell the user it was created

Folder exist:

Yes: ls / show files

No: mkdir / create folder

Lab 17

- Take a no. (n)
 - ↓
 - separate its digit (the integer)
 - ↓
 - square them
 - ↓
 - add those square
 - ↓
 - Replace the original no. with this new sum
 - ↓
 - Repeat
 - ↓
 - if reached 1 → Happy no.
 - ↓
 - if the no. start repeating in loop & never reaches 1 → Not happy

echo "enter a no."

```

n = $1
while [ $n -ne 1 ]
do
    sum = 0
    while [ $n -gt 0 ]
    do
        digit = $(n % 10)
        sum = $(($sum + digit * digit))
        n = $(($n / 10))
    done
    n = $sum
done
echo "happy number"
  
```